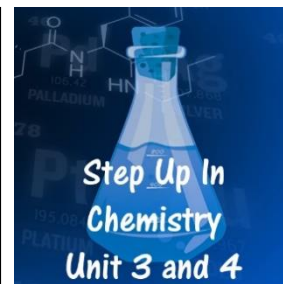
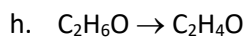
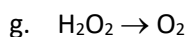
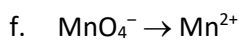
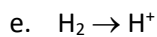
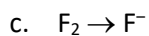
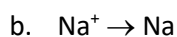
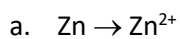


3.2 Redox Equations

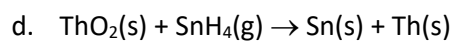
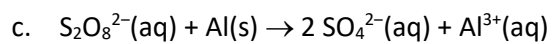
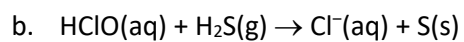
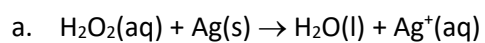
Problems Worksheet



1. Balance the following half equations and state whether the process is oxidation or reduction. Assume all processes are occurring in an acidic solution.



2. Write balanced redox equations for the following processes occurring in acidic solutions. Identify the oxidising agent in each of the reactions.

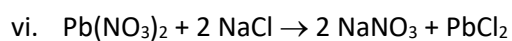
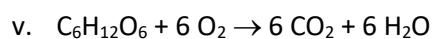
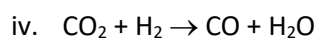
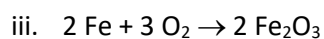
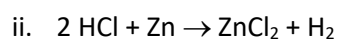
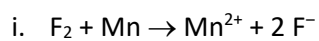


3. Write an equation for each of the following reactions and classify the process as metal displacement, halogen displacement, combustion or corrosion.
- Ethane (C_2H_6) is burnt in air.
 - A strip of aluminium is placed in a solution of copper(II) nitrate.
 - Chlorine gas is bubbled through a solution of potassium iodide.
 - Samarium metal forms Sm_2O_3 when exposed to air.

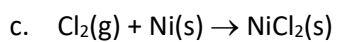
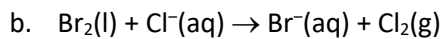
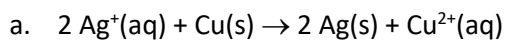
4. An oxidising agent and a reducing agent can be identified in redox reactions.

a. Explain what is meant by an oxidising agent and a reducing agent.

b. Identify the oxidising and reducing agents in each of these reactions:



5. Use the table of standard reduction potentials to determine whether the following reaction will proceed:



d. A piece of zinc is dropped into a jar of fluorine gas.

e. A solution of chromium(III) nitrate is poured over a strip of tin.

f. Acidified dichromate solution is mixed with hydrogen peroxide solution.

6. Displacement reactions are examples of oxidation and reduction.
- Describe the relationship between the activity series of metals and the position of the metals on the table of standard reduction potentials.
 - A strip of aluminium is dropped into a solution of nickel(II) nitrate. Write an equation and an observation for the reaction.
 - What is the overall voltage of this reaction? What does the voltage tell us about the spontaneity of the process?
 - Describe what would be observed when a strip of nickel is dropped into a solution of aluminium nitrate.
 - Explain why fluorine gas cannot be easily produced by a displacement reaction with another halogen.